



FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGIES

SUMMER 2026

FINAL EXAMINATION

COURSE TITLE:	Compiler Construction
COURSE CODE:	CS4110
CREDIT VALUE	6
INSTRUCTOR:	Engr. Tanwi Nkiamboh
DEPARTMENT	ICT
EXAMINATION HALL	B.S. Chumbow Hall
DATE:	Tuesday, September 29, 2026
TIME:	14:00 - 16:30

INSTRUCTIONS:

Each student will present in 3 minutes.

Your report should not exceed 30 pages

Attempt ALL questions

Lexical and Syntactic Analysis of Informal Urban Communication in Yaoundé

Project Context:

Yaoundé is a multilingual city where people communicate using a mixture of English, French, Pidgin, Fulfulde, Ewondo expressions, slang (“franc-anglais”), and city-specific abbreviations. Students must collect real conversations from taxis, chop houses, roadside businesses, neighborhoods, bike riders (bendskins), and the University environment, etc.

Project Objective



To design and implement a mini-language analyzer capable of performing lexical and syntactic analysis on real spoken expressions from Yaoundé, and to detect patterns related to commuting, market activities, security issues, weather, transport, electricity, and everyday transactions.

Project Components

1. Data Collection (Unique to Each Student) (10 marks)

Divide yourself in groups of 3. Each group must record **10–15 real-life statements** heard in Yaoundé on topics such as:

- i. Taxi and commuting issues
- ii. Poor internet connectivity
- iii. Limited Electricity supply
- iv. Market bargaining
- v. Rainy season struggles
- vi. Fuel scarcity
- vii. Roadside business interactions
- viii. Bendskin communication
- ix. Security announcements or checkpoints
- x. Life at the ICT University

Students **MUST** *manually transcribe* these sentences. (That is listen carefully and **write the exact words** spoken — including slang, code-mixing, accents, incomplete sentences, or mistakes)

2. Lexical Analysis

1. Identify tokens such as
 - i. **Nouns (e.g., tchop, quartier, bendskin)**
 - ii. **Verbs (drop me, hala me small money, dey for front)**
 - iii. **Slang words (hmmm, garr, zéro-zéro, je wanda, ekiee)**
 - iv. **Code-mixed expressions** (French + English + Pidgin) (10 marks)
2. Create a **custom lexical specification** using:
 - i. Regular expressions for token types
 - ii. A small LEX/FLEX program **or** a custom lexical analyzer in any programming language of your choice (Java/Python)
3. Analyze token frequency and variation. (10 marks)

3. Syntactic Analysis

1. Construct a **context-free grammar (CFG)** that describes the structure of the expressions collected.



2. Show steps:
 - i. Remove left recursion
 - ii. Do left factoring
 - iii. Compute **FIRST and FOLLOW** sets
 - iv. Build an **LL(1)** parsing table
 - v. Or construct **LR(0) or SLR(1)** items
3. Test their grammar using **their own collected sentences**.
4. Show which sentences are accepted or rejected.

(5, 5, 5, 5 marks)

4. Implementation Requirement

Implement:

- i. A **simple parser** (LL(1) or SLR(1))
- ii. That can read their tokenized inputs
- iii. And determine whether a sentence fits their constructed grammar

(5, 2, 3 marks)

5. Final Deliverables

A. Report (25–30 pages) (5 marks)

Must include:

- i. Raw collected statements
- ii. Token tables
- iii. Regular expressions
- iv. Grammar rules
- v. Parsing table or LR automaton
- vi. Screenshots of working analyzer
- vii. Discussion on *why Yaoundé communication is linguistically complex*

B. Source Code

- i. Lexical analyzer
- ii. Parser
- iii. Test cases taken from your data

C. 10-Minute Power Point Presentation and Demonstration (5 marks)

